

Uncertainty-Guided Edge Learning for Deep Image Regression in Remote Sensing

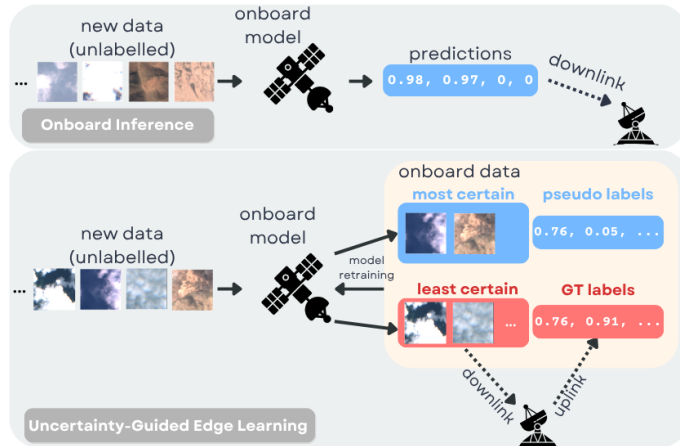
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ABSTRACT

Edge learning refers to training machine learning models on edge devices using newly collected on-board data. However, the limited computational resources of edge devices constrain both model optimisation and the estimation of predictive uncertainty on unlabelled data, which is essential for guiding model updates.

We propose an uncertainty-guided edge learning (UGEL) algorithm that prioritises data to accelerate convergence of onboard regression models. UGEL estimates predictive uncertainty using deep beta regression, where a neural network predicts the parameters of a beta distribution for the target value given an input image. Unlike existing uncertainty estimation methods that are computationally expensive on edge devices or rely on restrictive distributional assumptions, deep beta regression (DBR) requires only a single forward pass and supports more flexible predictive distributions. Results show that UGEL achieves faster convergence than active and semi-supervised learning methods.

DIAGRAM



(Top) Onboard inference for responsive remote sensing missions
(Bottom) The proposed UGEL for remote sensing missions

UGEL FRAMEWORK

Uncertainty estimation:

Let $h_{\mathbf{w}}$ be the uncertainty estimator, $\mathcal{I} = \{x_i\}_{i=1}^N$ be the unlabelled data. The predictive uncertainty of $f_{\mathbf{w}}$ on x_i is $h_{\mathbf{w}}(x_i)$.

Subset selection:

In each round of UGEL, $\mathcal{T} = \mathcal{T}_U \cup \mathcal{T}_C$ is composed from

A subset $\mathcal{T}_U \subset \mathcal{I}$ of size B_U contains the *most uncertain* samples.

$$\mathcal{T}_U = \arg \max_{S \subset \mathcal{I}, |S|=B_U} \sum_{x_i \in S} h_{\mathbf{w}}(x_i).$$

A subset $\mathcal{T}_C \subset \mathcal{I}$ of size B_C contains the *most certain* samples.

$$\mathcal{T}_C = \arg \min_{S \subset \mathcal{I}, |S|=B_C} \sum_{x_i \in S} h_{\mathbf{w}}(x_i).$$

Model updating:

Model updating combines supervised loss on human-labelled samples with semi-supervised loss on pseudo-labelled samples.

UNCERTAINTY ESTIMATION WITH DBR

Instead of predicting $y \in [0, 1]$, $f_{\mathbf{w}}$ is trained to output the mean μ and precision ν of a beta distribution given $\mathcal{X}$.

Differential entropy is then calculated to measure the predictive uncertainty of the fitted beta distribution,

$$h_{\mathbf{w}}(x) = \log \frac{\Gamma(\hat{\mu}\hat{\nu})\Gamma((1-\hat{\mu})\hat{\nu})}{\Gamma(\hat{\nu})} + (\hat{\nu}-2)\psi(\hat{\nu}) - (\hat{\mu}\hat{\nu}-1)\psi(\hat{\mu}\hat{\nu}) - ((1-\hat{\mu})\hat{\nu}-1)\psi((1-\hat{\mu})\hat{\nu})$$

where $\Gamma(\cdot)$ is the gamma function and $\psi(\cdot)$ is the digamma function.

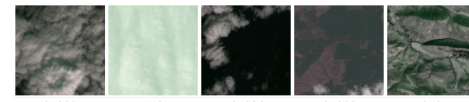
Higher entropy indicates that the model is more uncertain about the regression target value, whereas lower entropy reflects greater certainty.

DATASETS

Real scenes with real clouds patches are sourced from annotated Landsat-8 and Sentinel-2 cloud datasets.

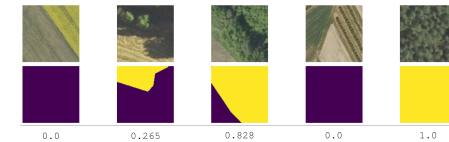
RSRC-L8: 20, 294 patches for learning and 22, 528 patches for testing.

RSRC-S2: 80, 000 patches for learning and 32,000 patches for testing.

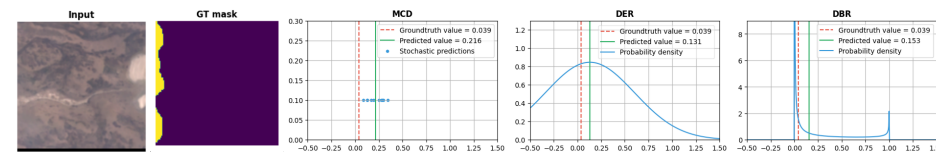


Real scenes with land coverage patches are sourced from Landcover.ai dataset.

RSRC-L8: 154, 056 patches for learning and 25, 279 patches for testing.



QUALITATIVE RESULTS



Qualitative comparison of Monte Carlo dropout (MCD), deep evidential regression (DER), and deep beta regression (DBR) in uncertainty estimation for cloud coverage prediction.

PSEUDOCODE

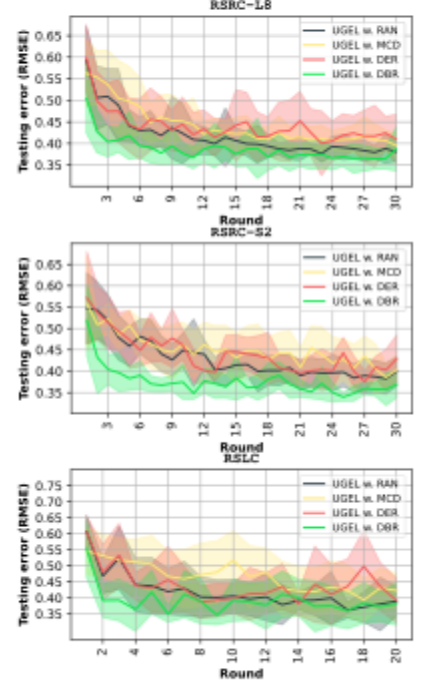
Algorithm 1 UGEL (single round)

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1: Inputs:
    $f_{\mathbf{w}}$ : regression model
    $h_{\mathbf{w}}$ : uncertainty estimator
    $f_{\mathbf{w}}$ : twin model
    $\mathcal{D} = \{(x_j, y_j)\}_{j=1}^M := (\mathcal{X}, \mathcal{Y})$ : init. labelled dataset
    $\mathcal{I} = \{x_i\}_{i=1}^N$ : unlabelled data (typ.  $N \gg M$ )
    $B_U$ : #samples for human annot. (req.  $B_U \leq N$ )
    $B_C$ : #samples for model annot. (req.  $B_C \leq N - B_U$ )
    $\tau$ : loss coefficient
2: Select data subset
    $\mathcal{T}_U \leftarrow \arg \max_{S \subset \mathcal{I}, |S|=B_U} \sum_{x_i \in S} h_{\mathbf{w}}(x_i)$ 
    $\mathcal{T}_C \leftarrow \arg \min_{S \subset \mathcal{I}, |S|=B_C} \sum_{x_i \in S} h_{\mathbf{w}}(x_i)$ 
3: Label data subset
    $\mathcal{Y}_U \leftarrow$  manually annotate  $\mathcal{T}_U$ 
    $\mathcal{Y}_C \leftarrow f_{\mathbf{w}}(\mathcal{T}_C)$  // Pseudo-labels
    $\mathcal{Y}_C \leftarrow f_{\mathbf{w}}(\mathcal{T}_C)$  // Pseudo-labels
4: Retrain model(s)
    $(\mathcal{X}, \mathcal{Y}) \leftarrow (\mathcal{X}, \mathcal{Y}) \cup (\mathcal{T}_U, \mathcal{Y}_U)$ 
    $\hat{\mathcal{Y}} \leftarrow f_{\mathbf{w}}(\mathcal{X})$  // Prediction
    $\hat{\mathcal{Y}} \leftarrow f_{\mathbf{w}}(\mathcal{X})$  // Prediction
    $L \leftarrow \mathcal{L}_s(\hat{\mathcal{Y}}, \mathcal{Y}) + \mathcal{L}_s(\hat{\mathcal{Y}}, \mathcal{Y}) + \tau \mathcal{L}_{RMSE}(\hat{\mathcal{Y}} \cup \mathcal{Y}_C, \mathcal{Y} \cup \mathcal{Y}_C)$ 
5:  $f_{\mathbf{w}} \leftarrow \text{modelUpdate}(f_{\mathbf{w}}, L)$ 
6:  $f_{\mathbf{w}} \leftarrow \text{modelUpdate}(f_{\mathbf{w}}, L)$ 
7: Outputs:
    $f_{\mathbf{w}}$ : updated regression model
    $f_{\mathbf{w}}$ : updated twin model
    $\mathcal{D} \leftarrow (\mathcal{X}, \mathcal{Y})$ : expanded labelled dataset
    $\mathcal{I} \leftarrow \mathcal{I} \setminus \mathcal{T}_U$ : reduced unlabelled data

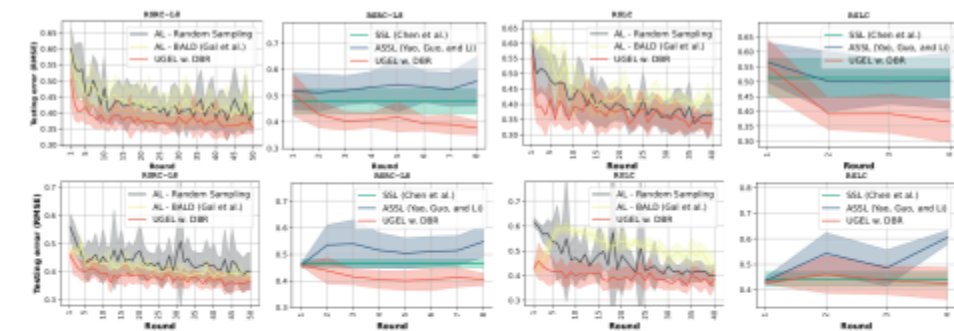
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ABLATION STUDIES



Comparison of different uncertainty estimation methods in UGEL on RSRC-L8, RSRC-S2, and RSRC-L8 using the ResNet18 backbone. Bolded curves and shaded regions respectively indicate mean RMSE testing error and std dev over 10 runs.

BENCHMARKING



Qualitative comparison of MC Dropout, DER and DBR in uncertainty estimation for cloud coverage prediction. The first two panels show the input image x and the foreground mask corresponding to the groundtruth coverage value y . The remaining panels show the three uncertainty estimation methods. Note that the predictive distribution of DER (a Gaussian) extends beyond the target range $[0, 1]$, while that of DBR (a Beta) lies within $[0, 1]$.



Paper



Code